GR4

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Our Previous Solution

“Reflection checkpoints given by AI before revealing solutions and adjustable levels of system involvement. “

Together, these support learning without forcing users into either full dependence or full independence, and allow assistance to adapt to different confidence levels and situations.

Our Pivoting Thought Process

Feedback:
Realizing our solutions could've been more creative and we should've clearly narrowed down our domain

Interview Analysis:
All of our interviewees viewed AI negatively regardless of using is a lot or not

Synthesis
We reorganized our groups to get a more concrete domain to create our new solution

What’s the Problem

- People of all backgrounds view AI advancements negatively
- AI gives too many shortcuts and discourages critical thinking
- Companies are using AI to take shortcuts for profit
- Human thought is being devalues

Our new solution

“Instead of giving answers after a student asks a question, the AI withholds solutions entirely unless the student earns them through demonstrated thinking.”

Project Name given our new solution: ThoughtAI

Project Name

ThoughtAI

Given our new solution, we've come up with thought AI. An app that functions similarly as chat forms of AI with the caveat that the AI doesn't reveal the solution until users show thought development

Value Proposition:

ThoughtAI for critical thinking because critical thinking is critical

Primary User: University Students

University students often rely on AI for homework and don’t fully learn the material they are supposed to be learning. When it comes time to use what they learn, many students find themselves at a loss

Kialo Edu

- Organizes debates into clear pro and con argument maps instead of messy comment threads
- What works for our problem space:
    - -Helps users break compex questions into structured, logical points
    - -Encourages critical thinking and civil discussion rather than quick reactions
    - -Let’s people explore multiple perspectives visually and see how arguments connect
- What doesn’t work:
    - -Is not a form of AI

Photomath

What works for our problem space:

- Provides fast access to correct answers, which helps users quickly unblock themselves when stuck
- Step-by-step solutions can clarify procedural math steps and reduce surface-level confusion
- Useful as a verification tool for checking final answers or identifying calculation mistakes

What doesn’t work:

- Encourages shortcut behavior that bypasses reflection and independent reasoning
- Gives users full solutions without requiring them to articulate their own thinking
- Lacks mechanisms to scaffold understanding based on user effort or partial reasoning
- Optimizes for correctness over learning, which conflicts with our learning-first progression model

Khanmigo

- Khanmigo is designed as an AI tutor that prioritizes learning over efficiency by encouraging students to reason through problems instead of immediately providing answers, which closely aligns with our goal of reducing shortcut driven AI use

- Its approach demonstrates that users are willing to engage with AI systems that intentionally slow down assistance in order to support understanding, validating our decision to withhold solutions until users demonstrate engagement

- Khanmigo primarily relies on conversational guidance and predefined tutoring flows, which limits how much control users have over pacing, depth of help, or when solutions become available

- Although Khanmigo supports explanation and reasoning, users can often continue prompting until they receive substantial guidance without clearly demonstrating sufficient understanding, reducing the sense of earned progress

- This creates an opportunity for our solution to differentiate by requiring users to actively demonstrate thinking in order to unlock deeper assistance, shifting agency more explicitly to the learner and reinforcing ownership of the problem solving process

Similarities

- Like our solution, these tools aim to help users engage with complex problems by breaking them into more understandable components rather than leaving users to face an unstructured or overwhelming task

- They all acknowledge that users benefit from seeing reasoning or structure, whether through argument maps, step by step explanations, or guided tutoring, reinforcing the importance of making thinking visible

- Each product supports users who are actively trying to make sense of a problem, not just passively consume information, aligning with our focus on learning oriented behavior

- They demonstrate that users are willing to interact with systems that slow down immediate answers in favor of explanation, structure, or guided exploration

- All of these tools attempt to support critical thinking in their own way, either by encouraging reasoning, comparing perspectives, or clarifying problem solving steps, which overlaps with our goal of promoting understanding over blind reliance

Differences

• Unlike these tools, our solution withholds the final solution entirely until users demonstrate sufficient understanding, whereas others eventually provide answers through continued prompting, scanning, or exploration

• Existing tools tend to optimize for explanation delivery or structured presentation, while our concept prioritizes earned access, making user effort a prerequisite rather than an optional behavior

• Photomath and similar tools allow users to bypass reflection by quickly accessing step by step solutions, whereas our system is explicitly designed to prevent shortcut behavior

• Khanmigo supports guided reasoning but does not strongly enforce user agency, allowing users to extract help without clearly proving understanding, which our solution treats as a core requirement

• Non AI tools like argument mapping platforms focus on organizing perspectives rather than adapting to user understanding, while our solution actively responds to the user’s demonstrated thinking to regulate access to help

Tasks

TASK 1 – SIMPLE

Student asks an initial question to receive guidance

User goal

Wants to ask an initial question to begin engaging with the problem and signal the need for help.

Why this task is simple

- Introductory and required for all users
- Requires only identifying a question or point of confusion
- No correctness, explanation, or justification required
- Serves as the entry point into the learning interaction

TASK 2 – MODERATE

Student answers guiding questions to progress through the problem

User goal

Wants to make progress by responding to the AI's guiding questions that walk them through the problem.

Why this task is moderate

- Requires articulating partial reasoning in response to prompts
- Involves committing to answers that may be incomplete or incorrect
- Requires sustained engagement across multiple steps
- Most users will complete this task to continue receiving guidance

TASK 3 – COMPLEX

Student solves a new problem independently to earn continued access to help

User goal

Wants to demonstrate full understanding by explaining their thought process and solving a new problem in order to unlock the ability to ask another question.

Why this task is complex

- Requires independently applying learned concepts to a new problem
- Requires explaining reasoning without step by step guidance
- High cognitive effort and sustained commitment
- Not all users will reach or complete this task
- Intended for advanced users seeking mastery rather than hints

Tasks

TASK 4 — Moderate-High

Student requests limited hints after an incorrect attempt and retries the problem

User goal

Wants to recover from an incorrect attempt by requesting a small number of hints and using them to try solving the problem again.

Why this task is moderate-high

- Requires recognizing that an initial attempt was insufficient
- Involves strategically requesting limited assistance rather than full guidance
- Requires applying hints to revise reasoning and retry the problem
- Demands sustained effort across multiple attempts
- Intended for users who are actively working toward understanding but still need targeted support

TASK 1 – SIMPLE

Student asks an initial question to receive guidance

User goal

Wants to ask an initial question to begin engaging with the problem and signal the need for help.

Why this task is simple

- Introductory and required for all users
- Requires only identifying a question or point of confusion
- No correctness, explanation, or justification required
- Serves as the entry point into the learning interaction

This task represents the student's first interaction with the learning system and prioritizes orientation over problem solving. At this stage, the student is trying to clarify what the problem is asking, surface initial confusion, and signal a need for guidance without pressure to perform correctly. The student's goal is simply to begin engaging with the material by asking an initial question, demonstrating awareness rather than mastery. Because this task does not require reasoning, explanation, or justification, it places a very low cognitive burden on the student and is accessible regardless of prior knowledge or confidence level. By removing the expectation of correctness, this task lowers anxiety and encourages honest engagement. It establishes a learning-first environment that invites curiosity and reflection while gently easing students into deeper, more demanding forms of reasoning later in the process.

Time Stamp in the video: 16s

TASK 2 – MODERATE

Student answers guiding questions to progress through the problem

User goal

Wants to make progress by responding to the AI's guiding questions that walk them through the problem.

Why this task is moderate

- Requires articulating partial reasoning in response to prompts
- Involves committing to answers that may be incomplete or incorrect
- Requires sustained engagement across multiple steps
- Most users will complete this task to continue receiving guidance

This task represents the stage where the student begins actively working through the problem with guidance. The student's goal is to make progress by answering guiding questions that help them think through the problem step by step. Unlike the initial task, this requires the student to explain parts of their reasoning and commit to answers, even if they are unsure or incomplete. This creates a moderate level of cognitive effort, as the student must reflect on their understanding and put their thoughts into words. The task is considered moderate because it encourages deeper engagement without requiring a full solution or perfect understanding. Most students will complete this step in order to continue receiving guidance, making it an important transition from initial orientation to more independent and meaningful problem solving.

Time Stamp in the video: 25s

TASK 3 – COMPLEX

Student solves a new problem independently to earn continued access to help

User goal

Wants to demonstrate full understanding by explaining their thought process and solving a new problem in order to unlock the ability to ask another question.

Why this task is complex

- Requires independently applying learned concepts to a new problem
- Requires explaining reasoning without step by step guidance
- High cognitive effort and sustained commitment
- Not all users will reach or complete this task
- Intended for advanced users seeking mastery rather than hints

This task represents the highest level of engagement within the learning experience and is designed for students who are ready to demonstrate genuine mastery. At this stage, the student is required to independently apply previously learned concepts to a new problem without receiving step by step guidance. The goal is not simply to reach a correct answer but to clearly explain their reasoning and problem solving approach, showing that they understand the underlying concepts rather than relying on prompts or hints. This task demands high cognitive effort and sustained commitment, as students must synthesize information, make decisions independently, and articulate their thinking without external scaffolding. Because asking another question is contingent on successfully completing this challenge, the stakes are higher and not all users will reach or complete this task. As a result, this task is appropriately classified as complex and is intended for advanced users who are seeking deep understanding and long term learning rather than quick answers. The AI assesses the student's explanation to determine whether they demonstrate sufficient understanding of the problem. Only after this understanding is demonstrated does the system allow the student to ask another question and continue engaging with the AI.

Time Stamp in the video: 46s

TASK 4 – Moderate High

User wants to compare their thinking with alternative perspectives

User goal

- Wants to see how other valid lines of reasoning differ from their own
- Wants to identify gaps, assumptions, or biases in their current understanding

Why this task is simple

Requires the user to evaluate and contrast multiple perspectives

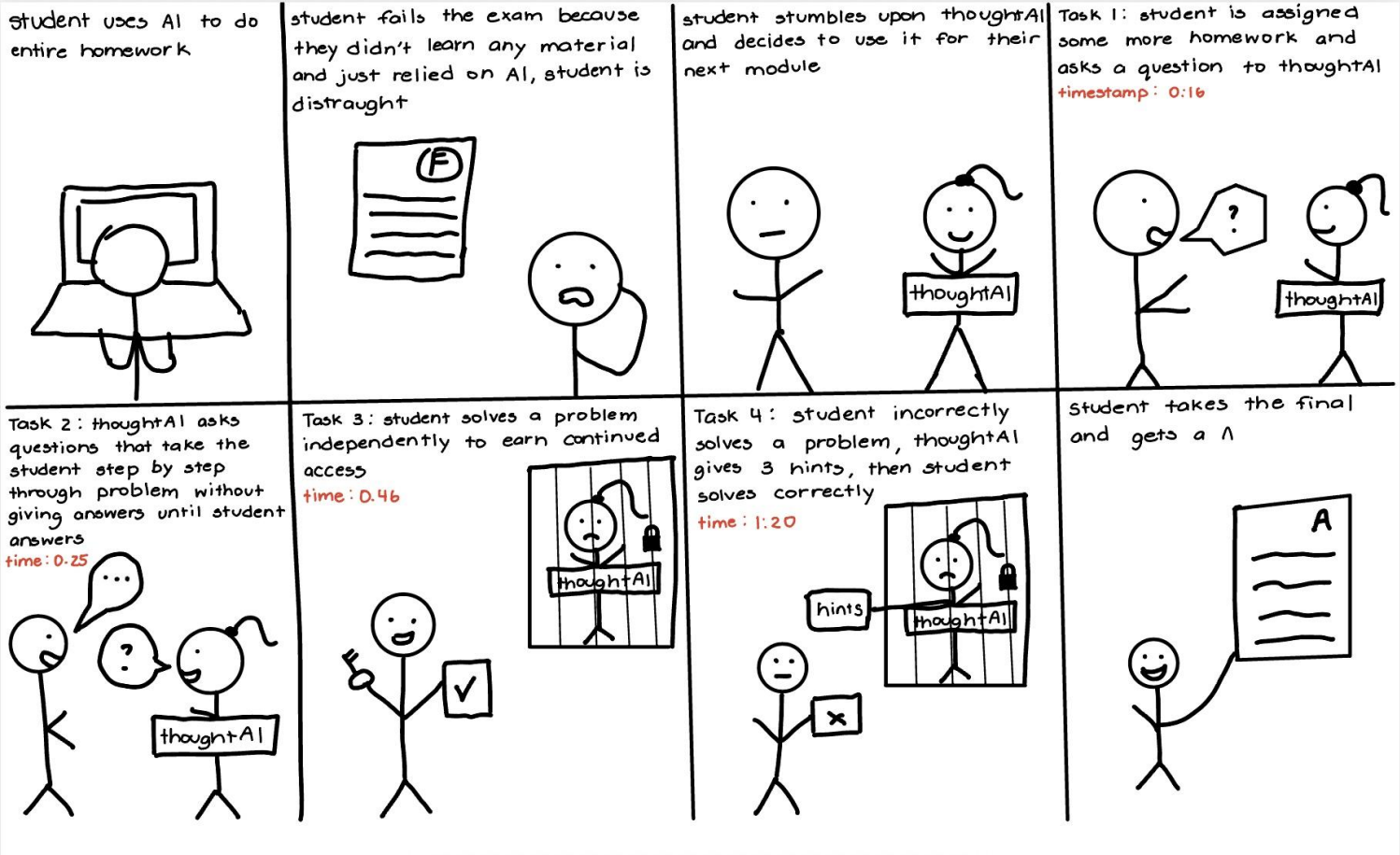
Involves reflection and judgment rather than simple explanation

Increases cognitive load but does not yet require full synthesis or justification

This task is designed for situations where a student attempts to solve a problem but does not demonstrate sufficient understanding on their first try. Instead of being shown the solution, the student is allowed to request a limited number of hints, specifically up to three, which are intended to guide them in the right direction without revealing the full answer. The student then uses these hints to revise their approach and attempt the problem again, reinforcing learning through iteration rather than correction. This task requires the student to recognize gaps in their understanding, selectively use guidance, and apply feedback across multiple attempts, making it more demanding than simply answering guided questions. Because assistance is constrained and success depends on how effectively the hints are used, this task is appropriately classified as moderate-high and supports learners who are actively working toward deeper understanding.

Time Stamp in the video: 1min20s

StoryBoard


student uses AI to do entire homework
student fails the exam because they didn't learn any material and just relied on AI, student is distraught
F
student stumbles upon thoughtAI and decides to use it for their next module
thoughtAI
Task 1: student is assigned some more homework and asks a question to thoughtAI
timestamp: 0:16
?
thoughtAI
Task 2: thoughtAI asks questions that take the student step by step through problem without giving answers until student answers
time: 0.25
...
?
thoughtAI
Task 3: student solves a problem independently to earn continued access
time: 0.46
thoughtAI
Task 4: student incorrectly solves a problem, thoughtAI gives 3 hints, then student solves correctly
time: 1:20
hints
thoughtAI
student takes the final and gets a A
A

Video Link:

https://youtube.com/shorts/eweid24nivl?feature=shared

Appendix

Design Pivot Context
This project reflects a pivot from an earlier AI assistance concept to a more narrowly defined learning focused solution based on feedback about domain clarity and differentiation.

Problem Framing Summary
The problem centers on growing negative perceptions of AI due to shortcut driven behavior that discourages critical thinking and devalues human thought.

Market Research Context
Comparisons to Kialo Edu, Photomath, and Khanmigo highlight existing approaches to reasoning support while revealing gaps in enforcing user effort before providing answers.

Task Progression Rationale
The tasks are intentionally structured to increase cognitive effort from asking a question to independently solving new problems in order to earn continued access to help.

Storyboard and Video Alignment
The storyboard maps directly to the final concept video and visually demonstrates each task through a clear emotional and behavioral progression.